IM08: TP7

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1 Delaunay filtering conditions

The filtering conditions specify how to obtain an α-shape from the Delaunay triangulation. To this end, we keep the triangles whose circumradius are smaller or equal than α. For each triangle, we are given its endpoints in $\mathbb{R}^3$. And, to obtain the circumradius, we can use the following formula for the area of a triangle:

$$A = \frac{abc}{4R}$$

where $a, b, c \in \mathbb{R}$ are the lengths of the sides of the triangle and R is the radius we are looking for. Thus,

$$R = \frac{abc}{4A}.$$

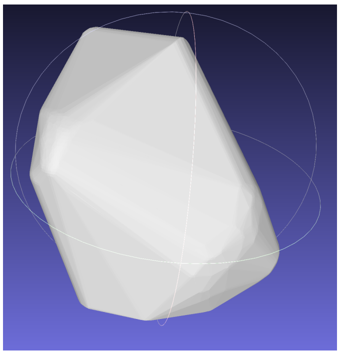
So, it suffices to compute a, b, c and A. To do so, we can use the norm of the vectors defined by the endpoints of the triangle, i.e.,

$$\begin{aligned} a &= \|y - z\|_2 \\ b &= \|x - z\|_2 \\ c &= \|x - y\|_2, \end{aligned}$$

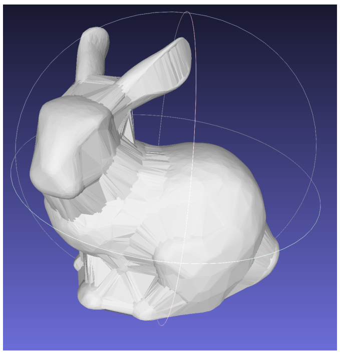
where $x, y, z \in \mathbb{R}^3$ are the endpoints of the triangle. After that, we can use a cross product in $\mathbb{R}^3$ to compute the area:

$$A = \frac{\|(y - x) \times (z - x)\|_2}{2}.$$

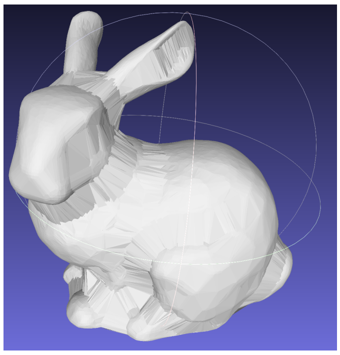
With that, we can finally compute R, and then filter out all triangles s.t. $R > \alpha$. The images below show some results for different alphas.



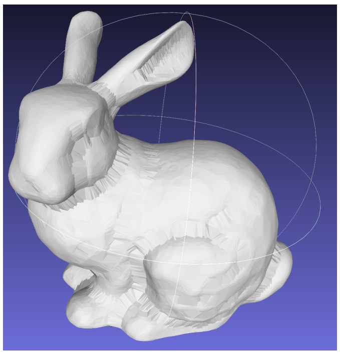
(a) $\alpha = 0.1$.



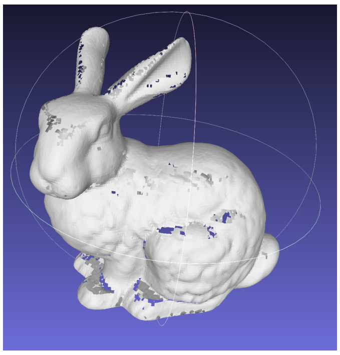
(b) $\alpha = 0.01$.



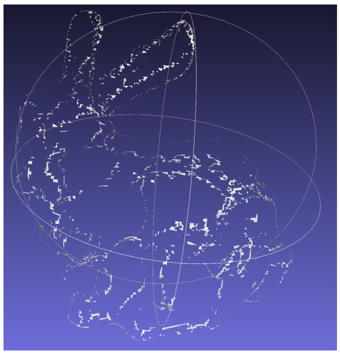
(c) $\alpha = 0.008$.



(d) $\alpha = 0.004$.



(e) $\alpha = 0.001$.



(f) $\alpha = 0.0008$.

Figure 1: Different α values for the filtering condition.